JUNJIE XIA

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SUMMARY

- Computational biologist with 5+ years of experience applying data science and machine learning to multi-modal biomarker discovery and developing scalable bioinformatics pipelines
- Specialized in analyzing and integrating single-cell, spatial transcriptomics, proteomics, and glycomics data to uncover disease mechanisms and therapeutic responses
- Skilled in Python, R, and workflow automation, bringing data-driven modeling and wet-lab collaboration to accelerate discovery of mechanism-based biomarkers

EDUCATION

PhD **University of Chicago** | Chicago, IL | Department of Chemistry — Sept 2021 – June 2026 (Expected)

Major courses: Chemical Biology, Fundamentals of Computational Biology, Machine Learning and Large-Scale Data Analysis

Honors: McCormick Fellowships (UChicago), Rising Stars in Measurement Sciences (2023, Merck), Olshansky Graduate Student Travel Award (UChicago)

BS **Zhejiang University** | Hangzhou, China | Department of Polymer Science — Sept 2017 – June 2021

Honors: National Scholarships for 2017-19 academic year (top 1% student)

RESEARCH EXPERIENCE

Institute for Genomics and Systems Biology, University of Chicago | Chicago, IL — Sept 2022 – Present

Graduate Researcher

- Led analysis of **multi-modal single-cell and spatial omics data (scRNA-seq, CITE-seq, ATAC-seq, Visium)** to identify biomarkers of immune cell differentiation
- Performed **multi-omics data integration** across transcriptomics, proteomics, and glycomics using framework such as **TotalVI and WNN**, comparing embedding results to that from single modality
- Designed and implemented reproducible workflows (Snakemake, Bash, and Python) for **data QC, alignment (STAR, Cell Ranger),** and downstream analysis using **Scanpy, Seurat, and clusterProfiler** across **multiple scRNA-seq platforms** (10x, Smart-seq2, Drop-seq)
- Applied **trajectory inference and RNA velocity analysis** (Monocle3, scVelo) and constructed **gene regulatory networks** (SCENIC, CellOracle) to elucidate transcriptional control of immune differentiation
- Built **machine learning models (Elastic Net regression)** to predict protein and glycan abundance from transcriptomic data and implemented **feature selection and explainability** modules for model interpretability
- Applied **multivariate statistical tests (MANOVA, PERMANOVA)** to evaluate differences across multi-omic signatures and assess condition-dependent biomarker shifts in large NGS datasets
- Visualized complex biological trends with custom Python/R scripts and **100+ publication-quality plots** for 5+ publications using **matplotlib, seaborn, and ggplot2,** including *Nature Methods* and *Cell* submission

Dept. Of Chemistry, University of Chicago | Chicago, IL — Sept 2021 – Sept 2022

Graduate Researcher

- Employed **non-compartmental PK models** in Python to simulate drug **ADME,** and parameterized models with experimental data to optimize nanoparticle formulation and dosing strategy
- Applied **survival analysis techniques** including Kaplan–Meier curves and log-rank tests to assess antitumor efficacy in preclinical mouse study

Dept. Of Polymer Science and Engineering, Zhejiang University | Hangzhou, China — Sept 2019 – June 2021

Undergraduate Researcher

- Conducted **Metropolis Monte Carlo simulation** to model polymer chain conformation and properties

TECHNICAL SKILLS

Programming & Cloud Computing: Python (PyTorch, scikit-learn), R, Bash, Git, SLURM HPC, AWS (EC2, S3)
NGS tools: STAR, Cell Ranger, Samtools, Scanpy, Seurat, variant calling pipelines
Data Types: NGS DNA/RNA-seq, single-cell & spatial transcriptomics (Visium, Xenium), proteomics (OLINK, MS)
Modeling & analysis: Bootstrapping, Monte Carlo, Principal component analysis, Regression modeling, Hypothesis testing, Bayesian methods, ML/AI modeling (Elastic Net, Random Forest, XGBoost)
Interest: Leverage data-driven insights to uncover biological knowledge for translational target & biomarker discovery

SELECT PUBLICATIONS (4 of 6)

Xia, J.[#], Phan, H.V.[#], Vistain, L.[#], Chen, M., Khan, A.A., Tay, S. Computational prediction of protein interactions in single cells by proximity sequencing. PLoS Comput Biol 20, e1011915 (2024).

Vistain, L., Keisham, B., **Xia, J.,** Phan, H.V. & Tay, S. Proximity sequencing for the detection of mRNA, extracellular proteins and extracellular protein complexes in single cells. Nat Protoc 19, 3568–3589 (2024).

Jiang, X., Lee, M., **Xia, J.,** Luo, T., Liu, J., Rodriguez, M., Lin, W. Two-Stage SN38 Release from a Core–Shell Nanoparticle Enhances Tumor Deposition and Antitumor Efficacy for Synergistic Combination with Immune Checkpoint Blockade. ACS Nano 16, 21417–21430 (2022).

Li, Y., Jiang, X., Luo, T., **Xia, J.,** Lee, M., Weichselbaum, R., Lin, W. TLR3 agonist nanoscale coordination polymer synergizes with immune checkpoint blockade for immunotherapy of cancer. Biomaterials 290, 121831 (2022).

MANUSCRIPTS IN SUBMISSION/REVIEW

Xia, J., Keisham, B., Pang, J. M., Padhi, A., Wang, H., Vasquez, M. G., Huang, L. M., Tay, S. Single-cell measurement of glycoforms, proteins and mRNA reveals dynamic glycan remodeling during T cell differentiation. *In revision (Nature Methods).*

Wang, H.[#], **Xia, J.[#],** Rahman, P.M.S.M.R., Keisham, B., Padhi, A., Deng, Y., Li, Y., Vistain, L., Kim, S., Vasquez, M. G., Khan, A.A., Clark, M.R., Tay, S. Spatial proximity sequencing maps developmental dynamics in germinal center. *In revision (Cell).*

Vasquez, M. G., Vizzard, V. J., Celiker, B., **Xia, J.,** Wang, G. A., Padhi, A., Tay, S. Accurate processing of ultra-short immune signals by single macrophages. *In revision (Science Advances).*

PATENTS & INTELLECTUAL PROPERTY

Tay, S., **Xia, J**. *Integrated Multi-Omics Profiling.* Provisional Patent Application No. 63/880,541, filed 2025.

CONFERENCE PRESENTATIONS

Poster, Merck-UChicago Lectureship, "Integrated profiling of proteins, glycoforms and mRNA in single cells reveals dynamic glycan remodeling during T cell differentiation." 2025

Poster, The Chicago BioEngineering Conference 2024, "Computational prediction of protein interactions in single cells by proximity sequencing." 2024

Poster, Merck Rising Stars in Measurement Sciences Symposium, "Spatial profiling of proteins, protein complexes and mRNA in human tonsil by proximity sequencing" 2023

LEADERSHIP AND SERVICE

Department of Chemistry, University of Chicago, Teaching Assistant 2021 – 2022

- Taught and supported over 50 undergraduate students in General Chemistry through weekly discussion sections
- Facilitated hands-on laboratory experiments and ensured proper safety and technique
- Held regular office hours to provide individual support and reinforce key concepts

PROFESSIONAL EXPERIENCE

Department of Computational Genomics, BridgeBio Pharma, Computational Genetics & Multiomic Extern Jan 2026 – Present

- Built an integrated tissue- and cell-specific PPI analysis pipeline combining tissue bulk proteomics, single-cell RNA expression, anatomy, spatial transcriptomics, and structural modeling to inform biomarker discovery
- Utilized AWS EC2 for scalable compute and AWS S3 for storage and data management to process, version, and analyze large public omics datasets, together with Git-based version control for code and analysis iteration
- Integrated multimodal publicly available datasets including tissue-specific PPI atlas, CellxGene Census sc-RNA data, HuBMAP anatomical and spatial resources, to contextualize PPIs across tissues and cell types

Department of Biomedicine and Healthcare, Kaitai Capital, Venture Capital Analysis Intern June 2021– Sept 2021

- Evaluated scientific and clinical data on cell-based therapies, including gene-edited CAR-T (e.g., PD-1 knockout), TCR-T for solid tumors, and gene therapies targeting CNS disorders
- Researched 15+ biotech companie and analyzed drug targets, mechanisms of action, and market competition; benchmarked nucleic acid aptamer-drug conjugates (ApDCs) against antibody-drug conjugates (ADCs)
- Supported 3 investment committee pitch decks and prepared 10+ internal review and due diligence memos; contributed to due diligence checklists, interview frameworks, and 10+ founder meeting summaries
- Developed internal tools in Excel and Python to streamline evaluation of clinical-stage assets, automating data extraction and risk stratification across therapeutic areas